REAR WHEEL BEARING SEAL INSTALLATION

AFFECTED VEHICLES: CERTAIN 2000 AND 2001 MODEL YEAR FOCUS

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NOTE: For disc brake equipped vehicles only, a spindle equipped with a double-lip water seal is supplied in the repair kit.

OVERVIEW

This document details the procedure for:

- **Drum brake equipped vehicles:** Installing ***new*** wheel bearings and water seal carriers onto both rear wheel hubs and installing a double-lip water seal onto both rear wheel spindles.
- **Disc brake equipped vehicles:** Installing ***new*** wheel bearings and water seal carriers onto both rear wheel hubs and installing ***new*** wheel spindles equipped with double-lip water seals.

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WHEEL HUB REMOVAL — ALL VEHICLES

NOTE: All figures in this document, except Figures 1 and 2, show rear drum brake configuration. Rear disc brake configuration is similar.

1. Raise the vehicle and remove both rear wheels.
   - **For vehicles with rear drum brakes:** Proceed to step 6.
   - **For vehicles with rear disc brakes:** Continue with step 2.
2. Remove the bolt securing the parking brake cable to the rear knuckle assembly, then disconnect the cable from the caliper. See Figure 1.

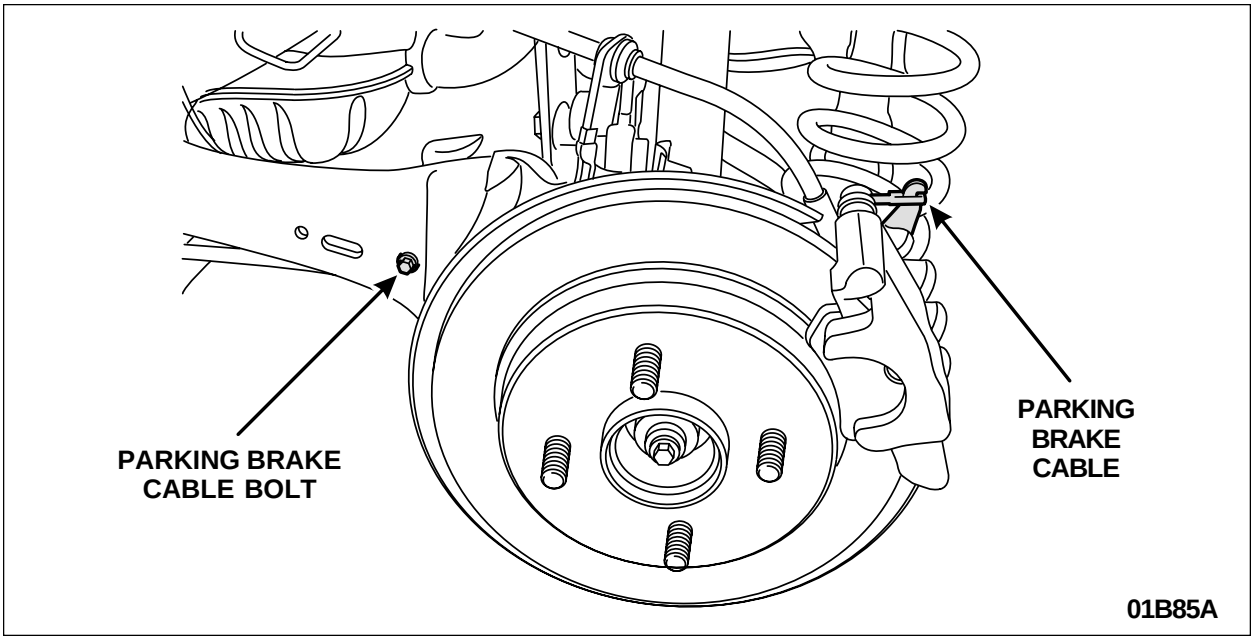


FIGURE 1

3. Remove the caliper bolts and position the caliper out of the way.
4. Remove the brake pads from the caliper bracket, then remove the caliper bracket.
5. Remove the brake rotor from the wheel hub.
6. Remove and discard the grease cap from the wheel hub assembly.

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7. **NOTE: As part of Safety Recall 00S31, vehicles were equipped with a retention cap which is secured to the left rear wheel spindle by a left-hand threaded bolt. The cap and bolt should be removed and discarded.**

   If equipped, remove the redundant wheel retention cap and bolt (installed as part of Recall 00S31) and discard. See Figure 2.

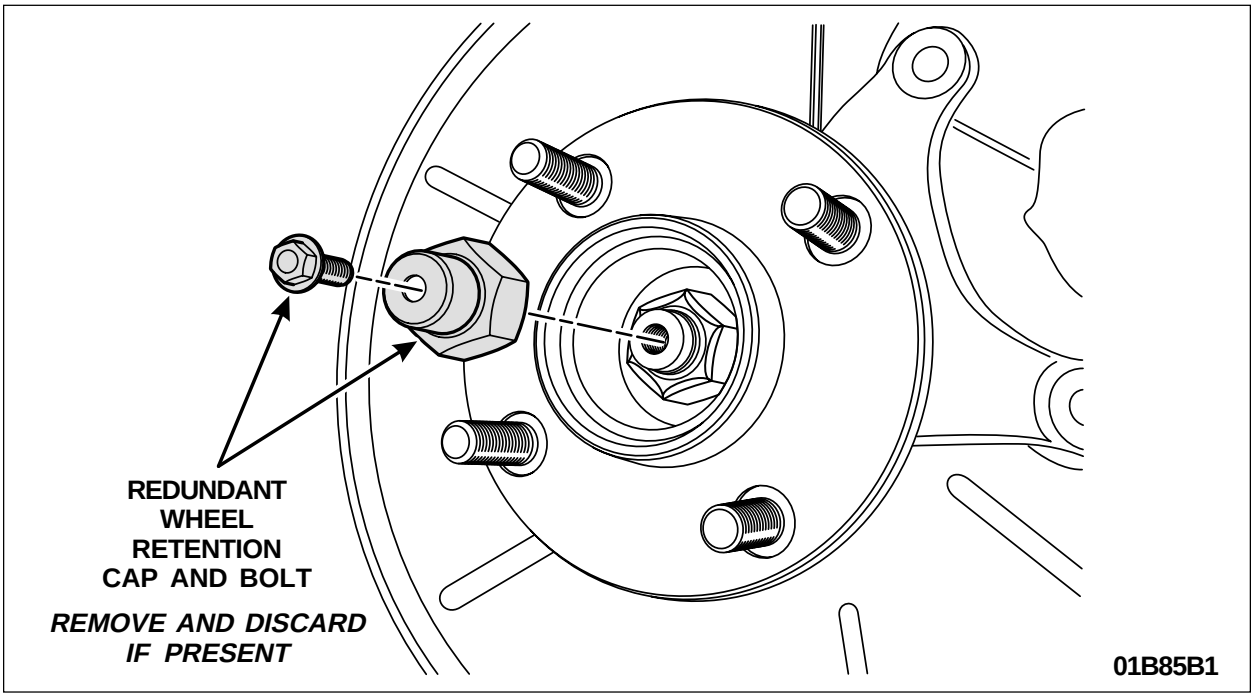


FIGURE 2

8. Remove and discard the wheel bearing retaining nut.
9. NOTE: For vehicles with rear drum brakes, tension in the parking brake system may need to be released to remove the wheel hub assembly.

   Remove the wheel hub assembly from the vehicle.

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HUB CLEANING/INSPECTION AND WHEEL BEARING REPLACEMENT — ALL VEHICLES

1. If equipped, remove and discard the ABS sensor ring from the wheel hub assembly. See Figure 3.

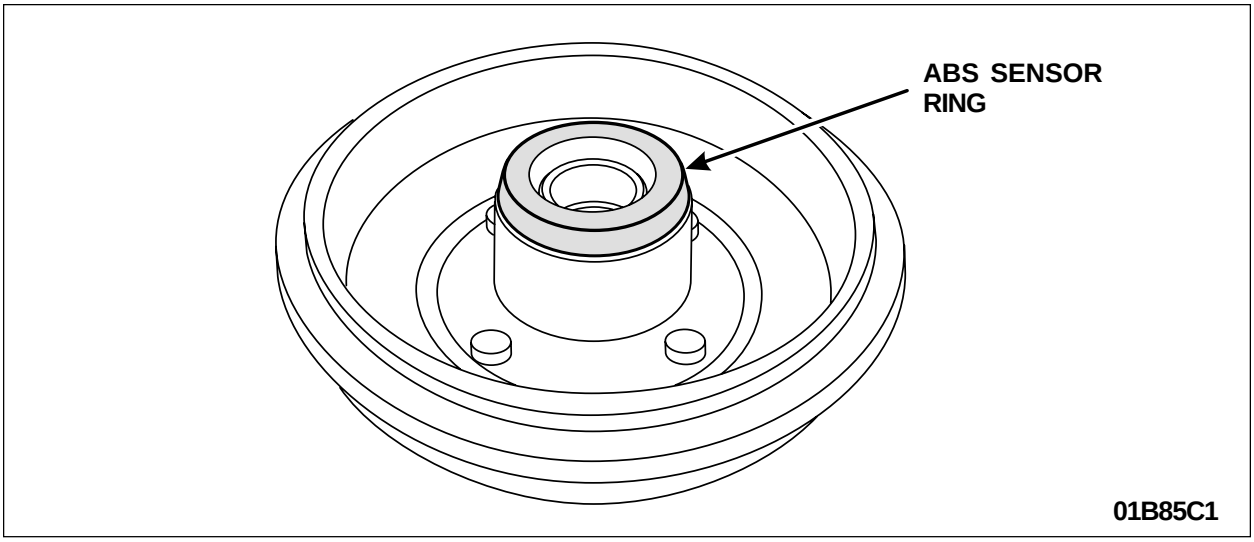


FIGURE 3

2. Remove the circlip that retains the wheel bearing in the hub assembly. Retain the circlip for reinstallation.
3. Using a suitable drift and press, press the wheel bearing out of the hub assembly.
4. NOTE: If the hub is found to be excessively pitted, even after cleaning, it must be replaced.

   Using the supplied 300 grit emery cloth, remove any corrosion that may exist in the wheel bearing area of the wheel hub assembly. See Figure 4.

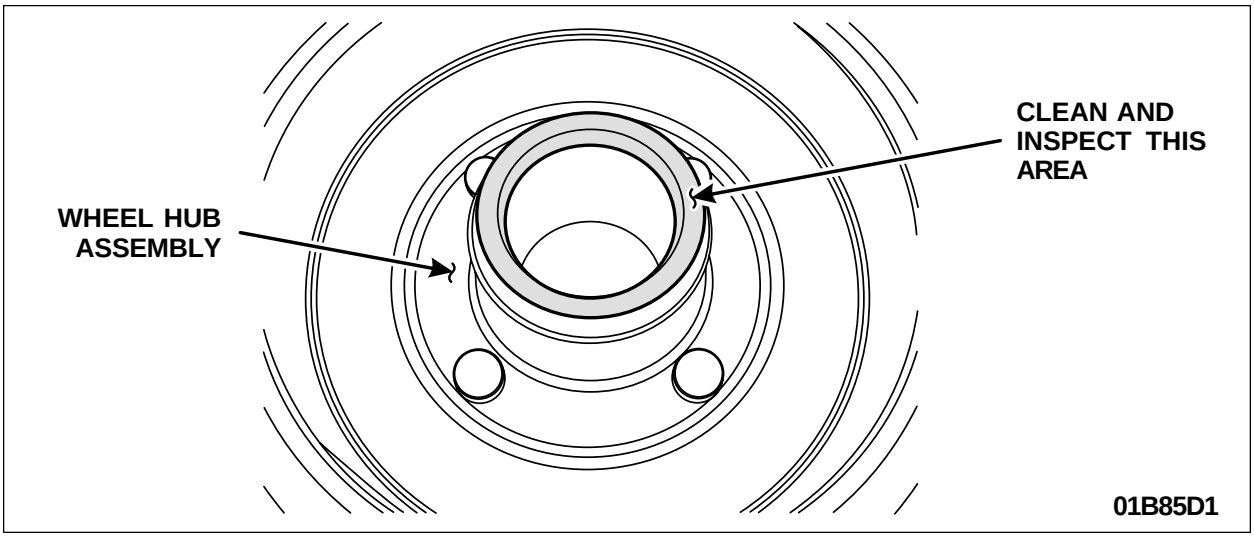


FIGURE 4

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5. **CAUTION: When installing the *new* wheel bearing into the hub assembly, be sure to apply the force to the outer race of the bearing.**

   NOTE: When installing the ***new*** wheel bearing into the hub assembly, the black rubber seal faces up.

   Using a suitable drift and press, press the ***new*** wheel bearing into the hub assembly. See Figure 5.

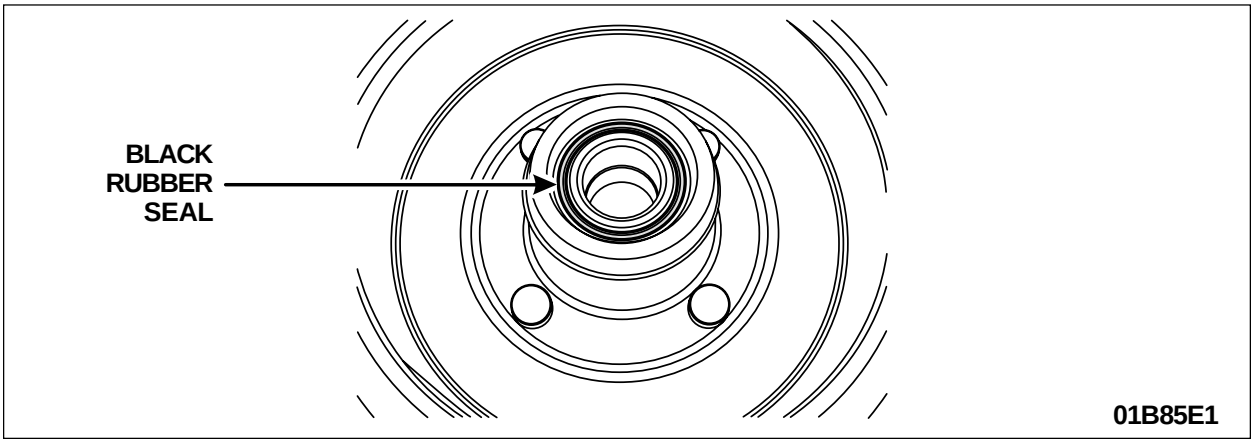


FIGURE 5

6. Install the circlip into the hub assembly.
7. Using Rotunda Tool Handle 205-153 combined with either Rotunda Water Seal Carrier Installer 204-322 for drum brakes or 204-323 for disc brakes, install the water seal carrier into the hub assembly. See Figure 6.

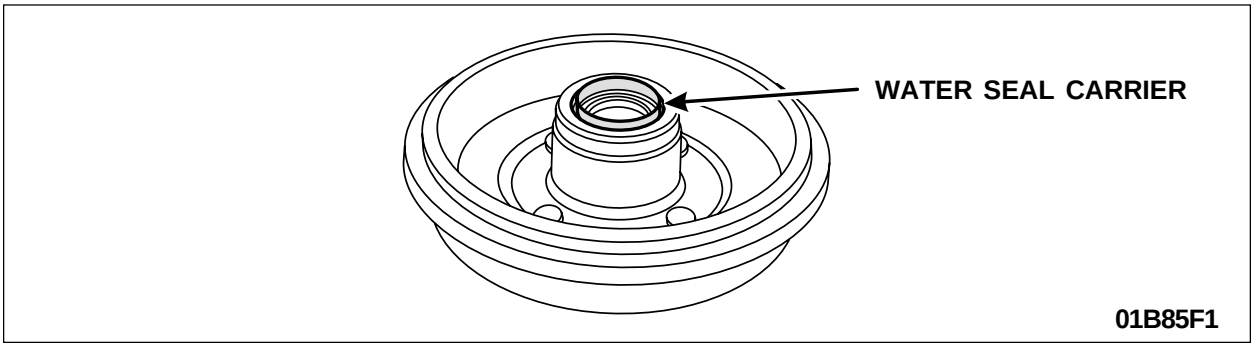


FIGURE 6

8. **CAUTION: The water seal carrier will protrude above the ABS tone ring when both are correctly installed. Use caution when installing the ABS tone ring to avoid damaging the carrier.**

   CAUTION: Use of a steel installation device may damage or affect the magnetic properties of the tone ring. Use only a non-ferrous device to install the tone ring.

   NOTE: Be sure to position the tone ring squarely onto the hub, then gently tap it into place using the proper tool.

   Using Installer 308-432 or equivalent non-ferrous installation device, position and gently tap the ***new*** ABS tone ring onto the hub, if equipped.

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SPINDLE PREPARATION/INSPECTION AND SEAL INSTALLATION — REAR DRUM BRAKE EQUIPPED VEHICLES ONLY

1. **CAUTION: The spindle is manufactured without a chamfer on the leading edge of the seal land, thus leaving a square, sharp corner. This corner must be removed, using the supplied emery cloth prior to seal installation, or damage to the seal will result.**

   NOTE: Cutting a chamfer into the leading edge is not required. Using the emery cloth to remove the sharp corner from the leading edge ensures the seal can be installed undamaged.

   CAUTION: The seal land must be sanded with the supplied emery cloth to remove any burrs and as much corrosion as possible.

   - **If the spindle is not pitted or corroded, the leading edge of the seal land must still be sanded or seal damage will result.**
   - **If the spindle is excessively pitted or corroded, even after sanding, it must be replaced.**

   NOTE: It is expected that a very small percentage (less than 3%) of spindles will require replacement due to corrosion/pitting concerns.

   Using the supplied 300 grit emery cloth, sand the spindle as described above. Remove any corrosion or burrs found on the wheel spindle. See Figure 7.

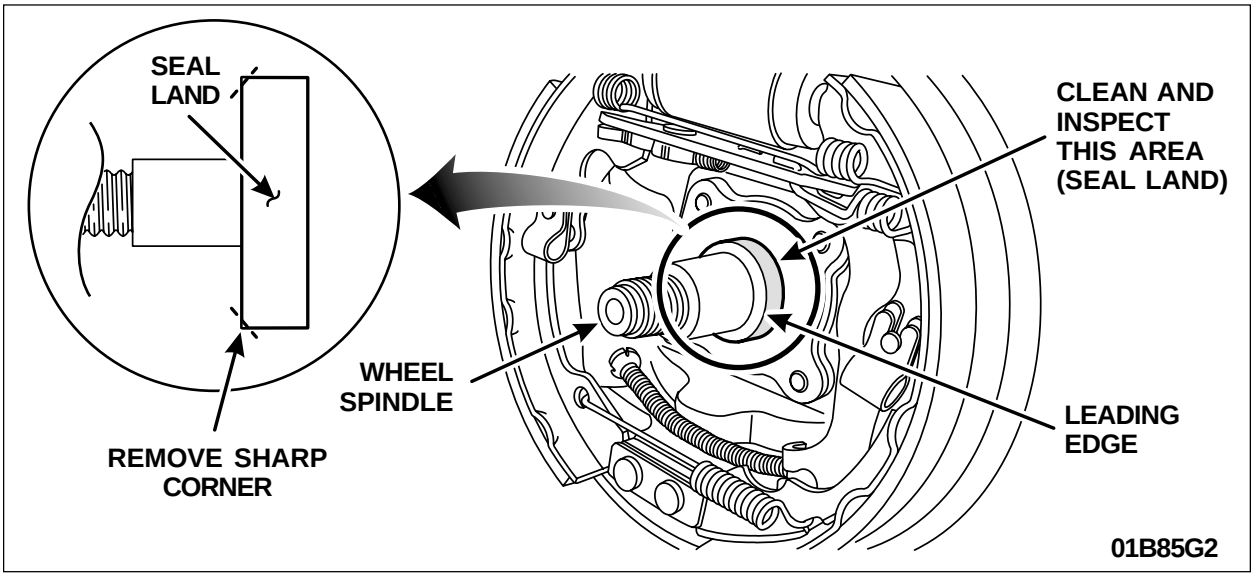


FIGURE 7

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2. **CAUTION: The double-lip water seal is prelubricated. Do not apply any lubrication to the double-lip water seal or the spindle before, during or after installation. Added lubrication may cause the double-lip water seal to unseat during vehicle operation.**

   NOTE: The double-lip water seal is oriented in the protective cap correctly for installation. Do not separate the double-lip water seal and cap until the seal is installed onto the wheel spindle.

   NOTE: Be sure to inspect the double lip water seal for any deformation (damage) after installation.

   Position the double-lip water seal and its protective cap onto the wheel spindle as far as possible by hand. Thread the Rotunda Seal Installer 204-325 onto the wheel spindle until it bottoms out on the protective cap. Next, using a 25 mm (1 in.) crowfoot wrench, turn the seal installer until it stops and then tighten it to 7 Nm (61 lb-in). Unscrew the Seal Installer from the wheel spindle and remove the protective cap by hand. After removing the protective cap, inspect the double-lip water seal to make sure it was not deformed during installation. If deformed, re-inspect the spindle to ensure the sharp corner was adequately removed, then obtain and install a replacement seal. See Figure 8.

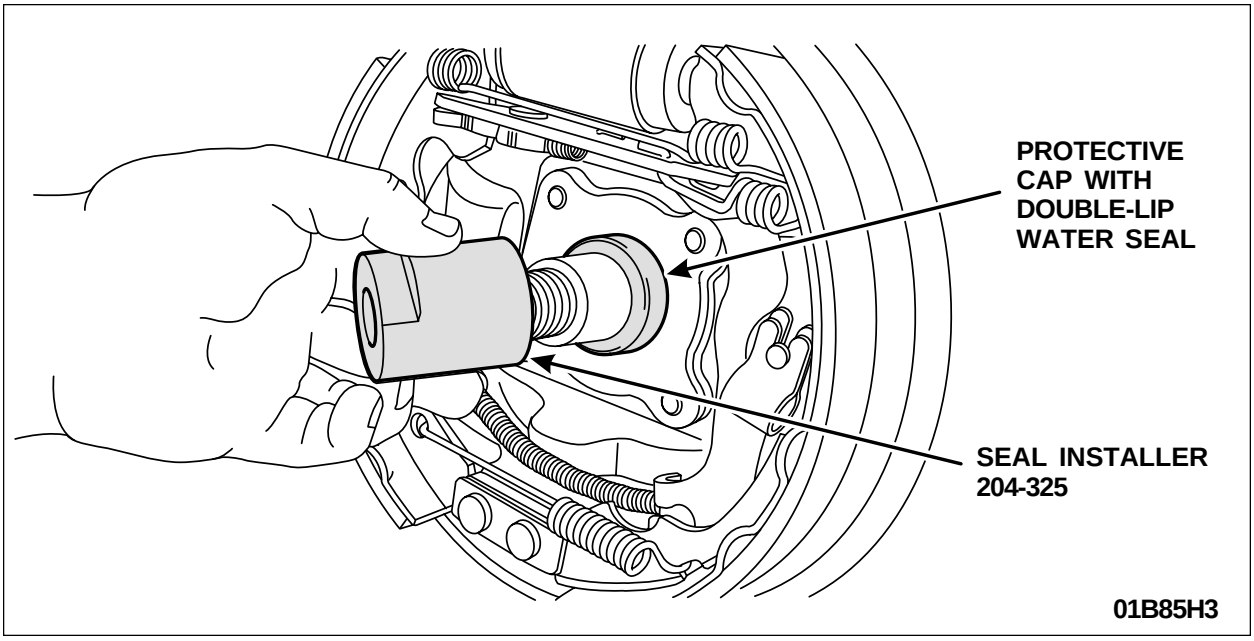


FIGURE 8

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SPINDLE REPLACEMENT — REAR DISC BRAKE EQUIPPED VEHICLES ONLY

1. Remove the anti-lock brake system (ABS) sensor and position it out of the way. See Figure 9.
2. Remove and discard the rear wheel spindle. See Figure 9. Retain the bolts for reassembly.

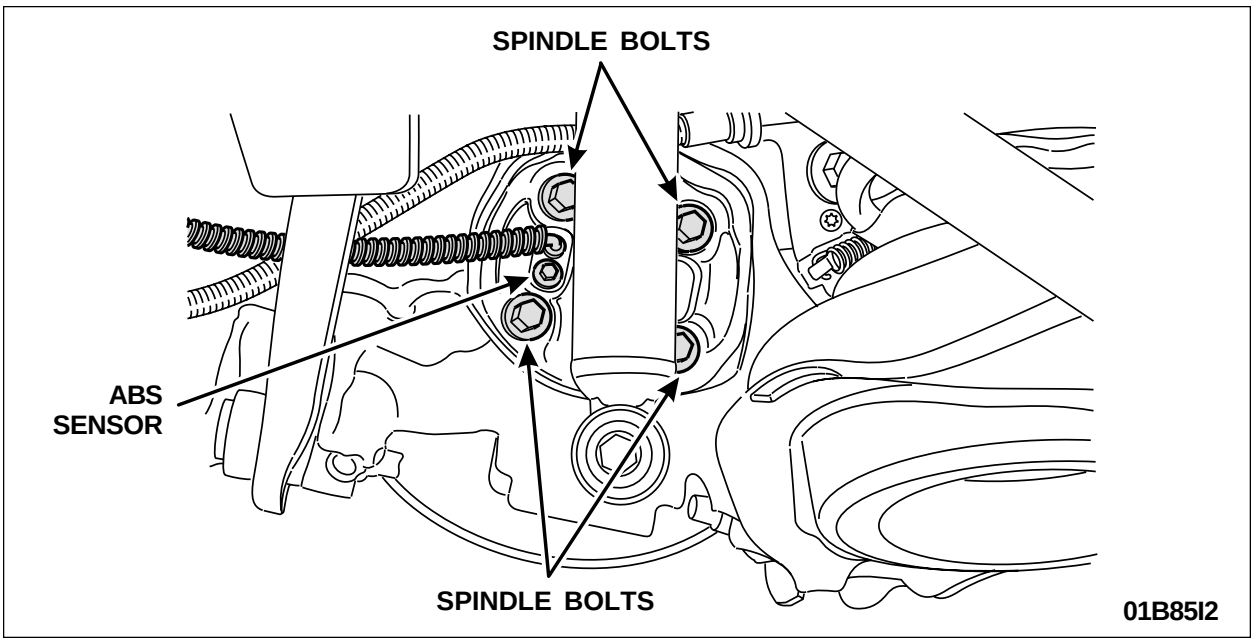


FIGURE 9

3. NOTE: Only the service parts kit for disc brake vehicles contains a ***new*** spindle. The ***new*** spindle comes with the double-lip water seal already installed.

   Install the ***new*** rear wheel spindle. Tighten the bolts to 66 Nm (49 lb-ft).
4. Position the ABS sensor and install the bolt. Tighten the bolt to 10 Nm (89 lb-in).

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WHEEL HUB INSTALLATION — ALL VEHICLES

1. **CAUTION: The double-lip water seal is prelubricated. Do not apply any lubrication to the double-lip water seal or spindle before, during or after installation. Added lubrication may cause the double-lip water seal to unseat during vehicle operation.**

   CAUTION: Be careful not to damage the *new* ABS sensor ring during installation.

   NOTE: Make sure the hub assembly is installed squarely and slowly to prevent any damage to the double-lip water seal.

   Install the wheel hub assembly onto the wheel spindle.

2. **CAUTION: The wheel bearing retaining nut must be tightened while turning the hub counterclockwise a minimum of ten (10) times.**

   Install the ***new*** wheel bearing retaining nut. Tighten the nut to 235 Nm (174 lb-ft).

3. Install the ***new*** grease cap onto the wheel hub assembly.
   - **For vehicles with rear drum brakes:** Beginning with Step 6 of Wheel Hub Removal — All Vehicles and ending with Step 3 of Wheel Hub Installation — All Vehicles, repeat all necessary service procedures in this document for the other rear wheel hub assembly. Once the other rear wheel hub assembly has been serviced, proceed to Step 9 of Wheel Hub Installation — All Vehicles.
   - **For vehicles with rear disc brakes:** Continue with Step 4 of Wheel Hub Installation — All Vehicles.
4. Position the brake rotor onto the wheel hub.
5. Install the caliper bracket and the brake pads. Tighten the caliper bracket bolts to 70 Nm (52 lb-ft).
6. Position the caliper onto the caliper bracket and install the caliper bolts. Tighten the bolts to 55 Nm (41 lb-ft).
7. Connect the parking brake cable to the caliper, then install the bolt securing the cable to the rear knuckle assembly.
8. Beginning with Step 2 of Wheel Hub Removal — All Vehicles and ending with Step 7 of Wheel Hub Installation — All Vehicles, repeat all necessary service procedures in this document for the other rear wheel hub assembly. Once the other rear wheel hub assembly has been serviced, proceed to Step 9 of Wheel Hub Installation — All Vehicles.
9. Install the rear wheels. Tighten lug nuts to 128 Nm (94 lb-ft).
10. If tension on the parking brake was released for wheel hub removal, the parking brake must be adjusted. Refer to Section 206-05 in the 2001 Focus Workshop Manual.

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